

CSS Phase 6: Responsive Design & Media Queries

What is Responsive Design?

Responsive Design হলো এমন একটি Design Technique যেখানে Website বিভিন্ন Screen Size-এ (Mobile, Tablet, Laptop, Desktop) সুন্দরভাবে কাজ করে।

একটি Responsive Website:

- ✓ Mobile-এ ভালো দেখায়
- ✓ Tablet-এ ভালো দেখায়
- ✓ Desktop-এ ভালো দেখায়
- ✓ Horizontal Scroll আসে না

Why Responsive Design?

বর্তমানে অধিকাংশ User Mobile Device ব্যবহার করে।

যদি Website Responsive না হয়:

- ✗ Text ছোট দেখাবে
- ✗ Layout ভেঙে যাবে
- ✗ User Experience খারাপ হবে
- ✗ SEO ক্ষতিগ্রস্ত হতে পারে

Viewport Meta Tag

Responsive Website-এর জন্য HTML-এ এটি অবশ্যই দিতে হবে।

```
""html id="y2x9v8" <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

এটি Browser-কে Screen Width অনুযায়ী Layout Adjust করতে সাহায্য করে।

Responsive Units

Percentage (%)

Parent Element-এর উপর নির্ভর করে।

```
```css id="a1g4v2"
width:50%;
```

---

## vw (Viewport Width)

```
```css id="j3z8p1" width:50vw;
```

50vw = Screen Width-এর 50%

vh (Viewport Height)

```
```css id="s5k7n3"
height:100vh;
```

100vh = পুরো Screen Height

---

## rem

Responsive Typography-এর জন্য Recommended।

```
```css id="m7x4q9" font-size:1.5rem;
```

Fixed Width vs Responsive Width

✗ Bad

```
```css id="w4t8b2"
```

```
.container{
 width:1200px;
}
```

✓ Better

```
```css id="e6y1c7" .container{ width:90%; }
```

✓ Best

```
```css id="r8n2f5"
.container{
 max-width:1200px;
 width:90%;
 margin:auto;
}
```

## Media Queries

Screen Size অনুযায়ী CSS পরিবর্তন করার জন্য Media Query ব্যবহার করা হয়।

Syntax:

```
```css id="u9h6m4" @media (max-width:768px){

}
```

Mobile Styles

Example:

```
```css id="p1v8z7"
h1{
 font-size:40px;
}
```

```
@media (max-width:768px){

 h1{
 font-size:24px;
 }

}
```

Desktop:

```
```text id="b4k3n9" 40px
```

```
Mobile:  
  
```text id="z7m1t5"  
24px
```

---

## Common Breakpoints

### Mobile

```
```css id="c8x4w1" @media (max-width:576px)
```

```
---  
  
### Tablet  
  
```css id="q2n7r6"  
@media (max-width:768px)
```

---

### Laptop

```
```css id="f5m9y3" @media (max-width:992px)
```

```
---  
  
### Desktop
```

```
```css id="t1v6p8"
@media (max-width:1200px)
```

---

## Responsive Flexbox

Desktop:

```
```css id="g7y2n4" .container{ display:flex; }
```

```
---

Mobile:

```css id="h3p8x5"
@media (max-width:768px){

 .container{
 flex-direction:column;
 }

}
```

Desktop:

```
```text id="r9c4m7" [1] [2] [3]
```

```
Mobile:

```text id="n6w2t1"
[1]
[2]
[3]
```

---

## Responsive Grid

Desktop:

```
```css id="k5q1v3" .grid{ display:grid;
```

```
grid-template-columns:
repeat(3, 1fr);
```

```
} ``
```

```
Mobile: css id="x8m6p2"
@media (max-width:768px){
  .grid{
    grid-template-columns:1fr;
  }
}
```

Auto Responsive Grid

Modern Technique:

```
``css id="d4y7k8" .grid{ display:grid;
```

```
grid-template-columns:
repeat(
  auto-fit,
  minmax(250px, 1fr)
);

gap:20px;
```

```
} `` Media Query ছাড়াই Responsive হবে।
```

Responsive Images

```
✗ Bad css id="j1m9c3"
img{
  width:800px;
}
```

✓ Good

```
```css id="s4v2n6" img{ max-width:100%; }
```

Image কখনো Container-এর বাইরে যাবে না।

---

## # Responsive Typography

```
```css id="f7p3w5"
h1{
  font-size:clamp(
    2rem,
    5vw,
    4rem
  );
}
```

Screen অনুযায়ী Font Size পরিবর্তন হবে।

Mobile First Design

Modern Development-এ Mobile First Approach Recommended।

প্রথমে Mobile Design:

```
```css id="v8n5r1" .card{ width:100%; }
```

পরে বড় Screen:

```
```css id="q6m2t4"
@media (min-width:768px){

  .card{
    width:50%;
  }

}
```

Container Pattern

সব Website-এ প্রায় ব্যবহার হয়।

```
```css id="w3x7k9" .container{ width:90%;
```

```
max-width:1200px;
```

```
margin:auto;
```

```
} ```
```

## Responsive Navbar Example

Desktop: `css id="n5p1v8"`

```
.nav{
```

```
display:flex;
```

```
justify-content:space-between;
```

```
}
```

Mobile:

```
```css id="k7m4x2" @media (max-width:768px){
```

```
.nav{
```

```
flex-direction:column;
```

```
}
```

```
}
```

```
---
```

```
# Most Important Responsive Properties
```

```
```css id="r2y8p4"
```

```
width:%
```

```
max-width
```

min-width

vw

vh

rem

clamp()

media query

flex-direction

grid-template-columns

---

## Phase 6 Practice Projects

### Beginner

1. Responsive Card
2. Responsive Image Gallery
3. Responsive Navigation

### Intermediate

1. Portfolio Website
2. Restaurant Website
3. Blog Website

### Advanced

1. E-commerce Homepage
2. Admin Dashboard
3. SaaS Landing Page

---

## Phase 6 Checklist

☐ Viewport Meta Tag

☐ %

- ☐ vw
- ☐ vh
- ☐ rem
- ☐ max-width
- ☐ Media Queries
- ☐ Mobile Breakpoints
- ☐ Responsive Flexbox
- ☐ Responsive Grid
- ☐ Auto-fit
- ☐ Minmax
- ☐ Responsive Images
- ☐ Clamp
- ☐ Mobile First Design
- ☐ Responsive Navbar
- ☐ Responsive Website Project

Responsive Design শিখে ফেললে তোমার Website Mobile, Tablet এবং Desktop—সব Screen-এ Professional দেখাবে। এটি Frontend Development-এর অন্যতম গুরুত্বপূর্ণ Skill।